

INTERNSHIP PROJECT REPORT

ABSTRACT

The Enterprise Network Configuration and Security Implementation Using Cisco Packet Tracer project focuses on designing, configuring, and analyzing a secure small-scale enterprise network using Cisco Packet Tracer. The primary objective of the project is to understand the fundamentals of computer networking, wireless communication, and basic cybersecurity practices through hands-on simulation.

The project involves configuring a wireless router, connecting wired and wireless client devices, assigning IP addresses using Dynamic Host Configuration Protocol (DHCP), and verifying network connectivity through various networking tools and commands. The simulated network enables communication between devices while demonstrating how routers, switches, IP addressing, and wireless technologies work together to provide reliable and secure network connectivity.

In addition to network configuration, the project emphasizes essential cybersecurity concepts such as secure wireless network configuration, authentication, firewall awareness, traffic monitoring, and protection against common network threats. Connectivity testing and troubleshooting techniques are performed to ensure proper communication between network devices and to identify potential configuration issues.

This project provides practical exposure to enterprise networking concepts, including IP addressing, routing, switching, packet forwarding, and wireless security. By using Cisco Packet Tracer, a realistic network environment is created where theoretical networking concepts can be implemented and analyzed without requiring physical networking equipment. The project enhances practical skills in network design, configuration, troubleshooting, and security, providing a strong foundation for further studies and careers in networking and cybersecurity.

MAPPING WITH SUSTAINABLE DEVELOPMENT GOALS

SDG GOAL	Reference to this Project
SDG 4 - Quality Education	• Provides hands-on learning in computer networking and cybersecurity. • Develops industry-relevant technical skills using Cisco Packet Tracer. • Bridges theoretical knowledge with practical implementation.
SDG 9 - Industry, Innovation and Infrastructure	• Builds resilient and reliable digital network infrastructure. • Promotes the use of modern networking technologies for enterprise environments. • Enhances technological capabilities through practical implementation of secure communication networks.
SDG 16 - Peace, Justice and Strong Institutions	• Promotes secure digital communication through network security practices. • Demonstrates cybersecurity measures such as authentication, firewall awareness, and secure wireless configuration. • Contributes to building trustworthy and secure information systems.

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LIST OF ABBREVIATIONS

Abbreviation - Full Form

IP - Internet Protocol

IPv4 - Internet Protocol version 4

DHCP - Dynamic Host Configuration Protocol

DNS - Domain Name System

SSID - Service Set Identifier

TTL - Time to Live

PC - Personal Computer

CMD - Command Prompt

MSSP - Managed Security Service Provider

IAM - Identity and Access Management

MDM - Mobile Device Management

ISO - International Organization for Standardization

CHAPTER 1

INTRODUCTION

Industrial and internship training forms a vital bridge between classroom learning and real-world professional practice. While academic coursework builds a strong theoretical foundation, it is through hands-on exposure in an actual working environment that a student truly understands how concepts are applied to solve practical problems.

This internship was undertaken with the aim of gaining first-hand experience in the field of cybersecurity, with a specific focus on networking, which forms the backbone of nearly every security operation carried out by an organization.

The internship was carried out at Raksha Technologies Private Limited, Chennai, a well-established cybersecurity solutions provider, over a period of three weeks from June 9 to June 30, 2026.

During this period, the intern was attached to the Networking Department, where the focus was on understanding how networking concepts such as routers, switches, firewalls, and IP addressing form the foundation upon which cybersecurity defenses are built. This report presents a detailed account of the objectives, the organization, the work carried out, the project undertaken, and the outcomes and reflections drawn from this internship experience.

CHAPTER 2

OBJECTIVES OF THE INTERNSHIP

The internship was undertaken with a set of clear objectives in mind, aimed at strengthening both technical understanding and professional readiness for a career in cybersecurity. The key objectives were as follows:

- To gain practical exposure to enterprise networking concepts including IP addressing, routing, switching, and wireless connectivity.
- To understand the role that networking infrastructure plays as the foundation of cybersecurity practices within an organization.
- To learn how devices such as routers, switches, and firewalls are configured, monitored, and secured in a real or simulated environment.
- To gain awareness of common network-based attacks, testing methodologies, and the tools used to detect and mitigate them.
- To develop hands-on skills using network simulation tools such as Cisco Packet Tracer for designing and testing secure network architectures.
- To understand the broader career landscape within cybersecurity and identify areas for continued self-improvement and learning.
- To build professional and interpersonal skills such as workplace communication, teamwork, and confidence in a new environment.

CHAPTER 3

COMPANY DETAILS

Company Name: Raksha Technologies Private Limited

Phone: 044-4345-5200

Industry: Cybersecurity Services

Location: 1st floor, Sree Narayana Complex, No 29, C' wing, Sarathy Nagar, Velachery, Chennai, Tamil Nadu 600042

Founded: 2000

About the Company:

Raksha Technologies Private Limited is a **Chennai-headquartered cybersecurity solutions provider** that has been operating in the information security space since 2000. It maintains additional branch offices in Coimbatore and Bengaluru, with partnered operations extending across the rest of the country.

The company positions itself as a pure-play cybersecurity integrator, offering end-to-end solutions that span *Cloud Security, Data Security, Web Security, Email Security, Compliance and Governance, Identity and Access Management (IAM) / Mobile Device Management (MDM), Application and Database Security, Network Security, Endpoint Security, and Security Operations*, in addition to **Managed Security Service Provider (MSSP)** offerings.

Raksha Technologies is an **ISO 9001** and **ISO 27001** certified organization, has received over 80 awards and recognitions from technology vendors and trade publications, and holds more than 20 brand partnerships with leading global security vendors.

CHAPTER 4

DESCRIPTION OF WORK DONE

The internship was carried out over a period of three weeks, from **June 9 to June 30, 2026**, within the Networking Department of Raksha Technologies. The placement in this department was intentional, as networking forms the foundation on which the majority of cybersecurity defenses, monitoring systems, and incident response strategies are built.

Throughout the internship, guidance and mentorship were provided by **Mr. Shailesh Kiran B**, *Senior Security Engineer*, who served as the primary trainer, along with support from **Mr. Anthony Raj A**, *HR Manager*, who ensured a smooth onboarding process and helped in navigating the organizational environment during the initial days.

During the course of the internship, exposure was gained to a wide range of topics including the fundamentals of computer networks, the working of routers, switches, and firewalls, IP addressing schemes, and wireless network configuration.

In addition to networking fundamentals, the internship also introduced various categories of cyberattacks, the general methodology behind penetration testing and vulnerability assessment, and the tools commonly used within the industry to identify, test for, and mitigate network-based threats.

The training combined conceptual explanations with practical demonstrations, allowing theoretical knowledge to be reinforced through simulation-based exercises.

CHAPTER 5

PROJECT ACTIVITIES

As part of the internship, a project titled "**Enterprise Network Configuration and Security Implementation Using Cisco Packet Tracer**" was carried out to simulate and configure a secure small-scale enterprise network. The objective of the project was to gain practical knowledge of network design, device configuration, wireless communication, IP addressing, and basic cybersecurity concepts using Cisco Packet Tracer.

The project began by designing a network topology consisting of a *cable modem, home wireless router, office desktop computer, bedroom desktop computer, laptop, television, and an external server connected through the Internet*. Both wired and wireless connections were established to simulate a real-world home/office network environment.

The wireless router was configured with appropriate network parameters, including the SSID, wireless security settings, and DHCP service. The DHCP server was enabled to automatically assign IPv4 addresses, subnet masks, default gateway addresses, and DNS server information to all client devices, eliminating the need for manual IP configuration.

The office desktop computer was connected to the router using an Ethernet cable and successfully obtained its IP configuration through DHCP. The laptop was configured to connect securely to the wireless network using the configured wireless profile and authentication settings. Network connectivity between all connected devices was verified after successful association with the wireless access point.

After completing the network configuration, communication between devices and external resources was tested. The office PC successfully accessed the wireless router's web-based management interface, confirming proper network configuration and router accessibility. The bedroom PC successfully accessed the hosted web server through the simulated Internet, demonstrating end-to-end connectivity, proper DNS resolution, and successful packet forwarding across the network.

Throughout the project, various networking concepts such as *IP addressing, DHCP configuration, default gateway assignment, routing, switching, packet forwarding, DNS resolution, and wireless communication* were studied and implemented. The project also introduced fundamental cybersecurity concepts including secure wireless authentication, network access control, firewall awareness, secure device configuration, and basic network troubleshooting techniques.

The project provided practical exposure to enterprise networking principles by integrating theoretical knowledge with hands-on configuration and testing. Using Cisco Packet Tracer enabled the simulation of a realistic networking environment without physical hardware, thereby strengthening practical skills in network deployment, configuration, verification, troubleshooting, and security implementation.

Outcomes:



Figure 1: Initial enterprise network topology created in Cisco Packet Tracer showing the cable modem, wireless router, wired and wireless client devices, and Internet connectivity.



Figure 2: Packet flow simulation demonstrating communication between devices and successful packet forwarding across the network.

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address.....: FE80::2E0:A3FF:FEAB:DD83
    IPv6 Address.....: ::
    IPv4 Address.....: 192.168.0.2
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                           192.168.0.1
```

Figure 3: Office PC receiving IPv4 address, subnet mask, default gateway, and DNS server automatically through DHCP.

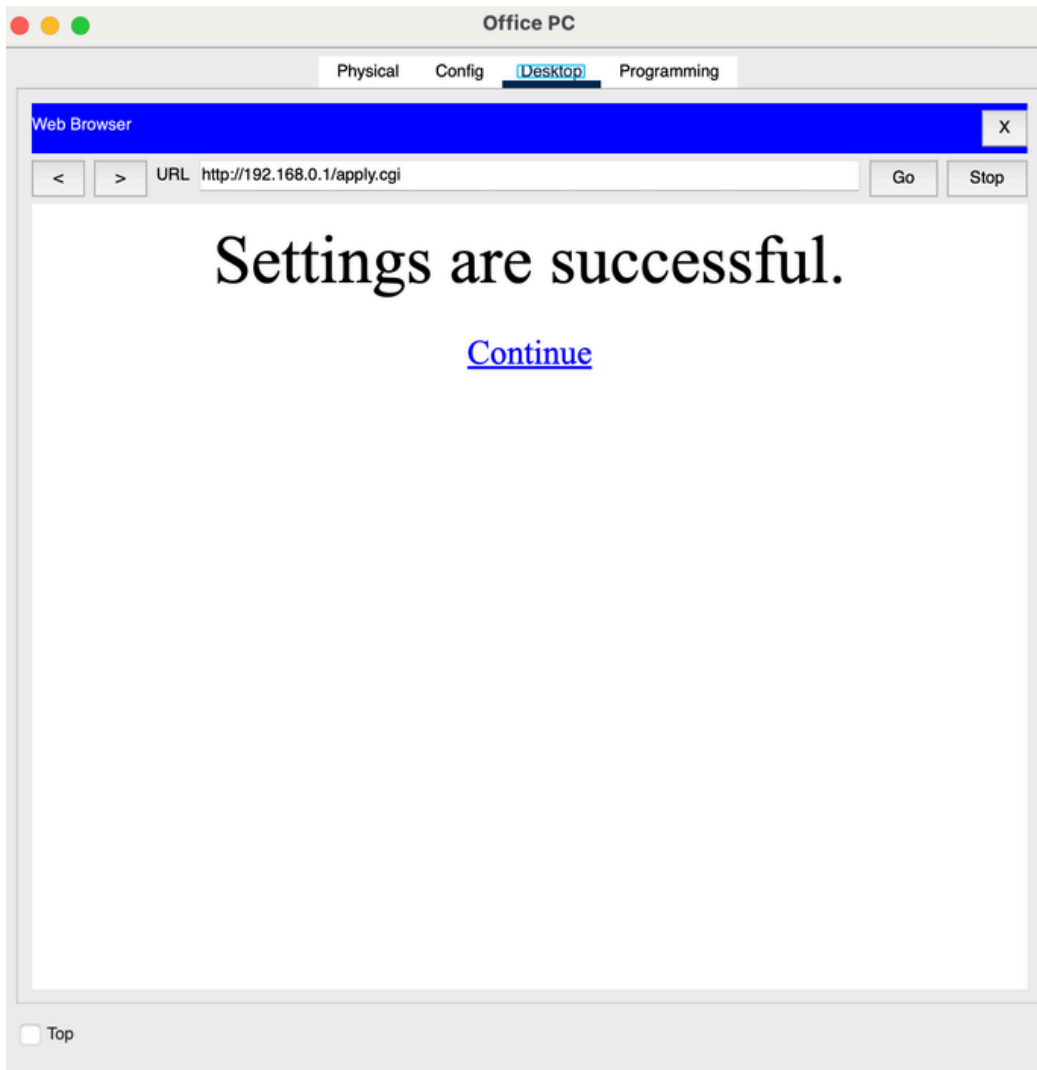


Figure 4: Successful application of wireless router configuration through the web-based management interface.



Figure 5: Laptop successfully connected to the wireless access point with strong signal strength and good link quality.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:

Reply from 192.168.0.1: bytes=32 time<1ms TTL=255
Reply from 192.168.0.1: bytes=32 time<1ms TTL=255
Reply from 192.168.0.1: bytes=32 time=21ms TTL=255
Reply from 192.168.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 21ms, Average = 5ms

C:\>ping skillsforall.srv

Pinging 209.165.200.232 with 32 bytes of data:

Reply from 209.165.200.232: bytes=32 time=22ms TTL=126
Reply from 209.165.200.232: bytes=32 time=3ms TTL=126
Reply from 209.165.200.232: bytes=32 time=2ms TTL=126
Reply from 209.165.200.232: bytes=32 time=2ms TTL=126

Ping statistics for 209.165.200.232:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 22ms, Average = 7ms
```

Figure 6: Connectivity verification using the Ping command, confirming successful communication between network devices.

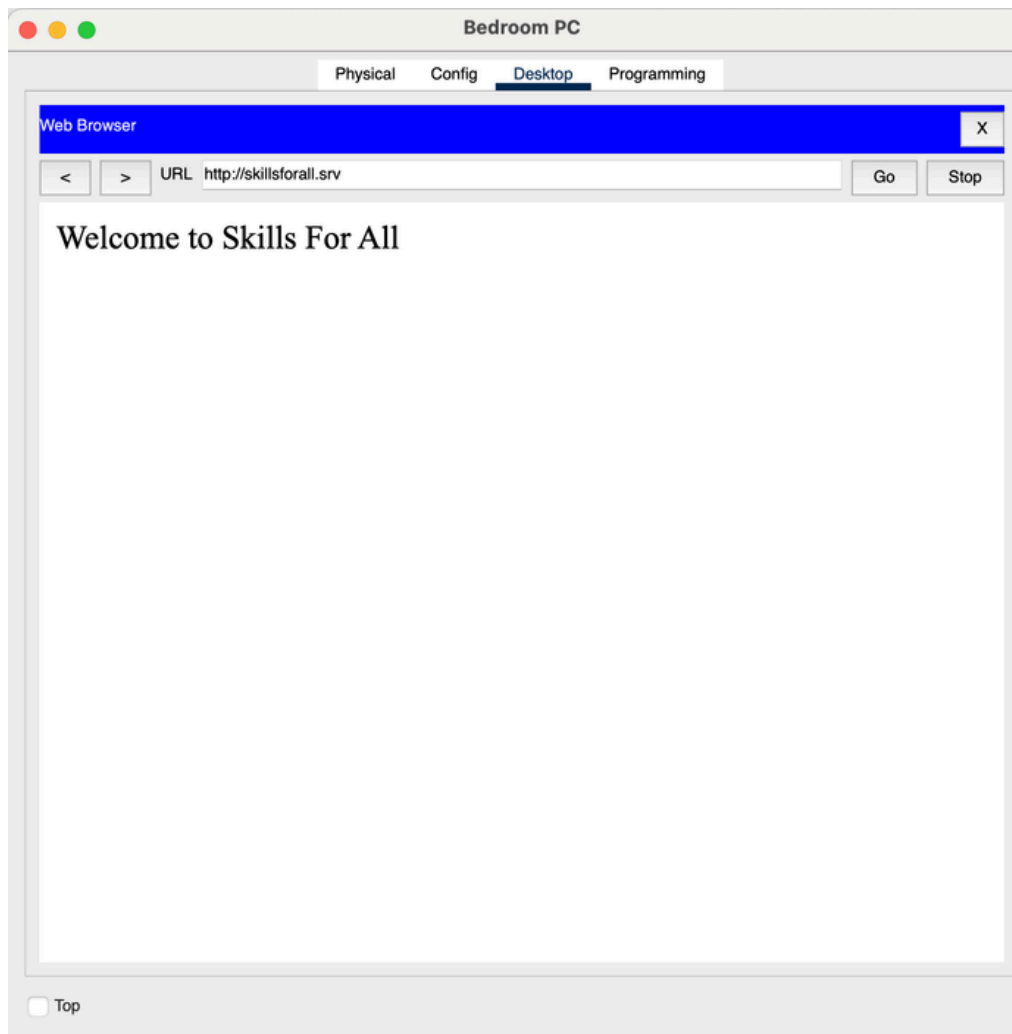


Figure 7: Bedroom PC successfully accessing the web server, verifying Internet connectivity and DNS resolution.

CHAPTER 6

LEARNING OUTCOMES

This internship provided substantial learning, both *technical* and *personal*.

On the technical front, a deeper understanding was gained of core networking concepts such as *routers, switches, firewalls, IP addressing, DHCP, and wireless network configuration*, along with an appreciation of how these elements come together to form the security backbone of an organization.

Exposure to various tools, common attack methodologies, and testing techniques provided a practical understanding of how cybersecurity operates in a real-world enterprise setting, well beyond what is typically covered in a classroom.

Beyond the technical aspects, the internship also offered valuable insight into the breadth of career paths that exist within the cybersecurity field, ranging from network security and security operations to compliance, cloud security, and beyond.

This exposure helped in understanding the importance of continuous self-evaluation and learning, given how rapidly the field of cybersecurity evolves. On a personal level, the internship helped build confidence in communicating within a professional environment and adapting to a new and unfamiliar setting, both of which are skills expected to be valuable throughout a future career.

CHAPTER 7

CHALLENGES FACED

At the very start of the internship, there was a degree of overconfidence stemming from prior academic knowledge of networking and cybersecurity concepts.

However, it soon became clear that there was a great deal more to learn, both in terms of the depth of technical detail expected in a professional setting and in terms of practical, hands-on application of these concepts.

This realization was an important turning point, encouraging a more open and humble approach to learning throughout the remainder of the internship, which continues even now. Another significant challenge was adjusting to a new professional environment.

Initially, there was hesitation in speaking up or interacting with colleagues, largely due to unfamiliarity with the workplace and the people in it. This led to a tendency to stay quiet rather than ask questions or participate actively in discussions.

Over the course of the internship, however, conscious effort was made to overcome this hesitation, gradually becoming more comfortable engaging with the team, asking questions when needed, and communicating more openly.

This was ultimately one of the more meaningful takeaways of the internship, alongside the technical learning.

CHAPTER 8

CONCLUSION

The *three-week internship* at **Raksha Technologies Private Limited** provided a valuable and well-rounded introduction to the field of cybersecurity, with a strong emphasis on networking as its foundation.

Working within the Networking Department, under the guidance of **Mr. Shailesh Kiran B, Senior Security Engineer** and with the support of the HR team, offered practical exposure to routers, switches, firewalls, and wireless network configuration, along with a broader understanding of network-based attacks and testing methodologies used in the industry.

The hands-on project involving Cisco Packet Tracer reinforced these concepts through simulation, allowing enterprise networking and security practices to be implemented and analyzed in a realistic, risk-free environment.

Beyond the technical learning, the internship also contributed meaningfully to personal growth, particularly in building confidence, improving communication in a professional setting, and adopting a more grounded, continuous-learning mindset.

Overall, this internship has provided a strong foundation for further academic study and future career pursuits in networking and cybersecurity.

INTERNSHIP ACCEPTANCE LETTER



ISO 9001:2015 | Cert : 727697
ISO/IEC 27001 : 2022 | Cert : 684037



Date: 25th May 2026

Dear Ms. Janavika,

Sub: Internship Acceptance Letter – Regd

We are pleased to inform you that you have been selected for an Internship at “Raksha Technologies Private Limited” at Chennai, India. We believe that you possess the necessary skills and qualities to contribute significantly to our organization.

The internship will commence on **9th June 2026** and will last till **30th June 2026**. During this time, you will have the opportunity to gain valuable hands-on experience from the work, and learn from our experienced team members. We are confident that this internship will provide you with a strong foundation for your future career endeavours.

As an Intern at “Raksha Technologies Private Limited”, you will be expected to adhere to our policies and guidelines, maintain a professional work ethic, and actively participate in the assigned tasks and projects. You will have access to resources and mentorship to support your growth and development throughout the internship period. We encourage you to take advantage of these opportunities and make the most of your time with us.

We believe that your skills, enthusiasm, and dedication will make a positive impact on our organization. We look forward to welcoming you to our team and providing you with a fulfilling and rewarding internship experience.

Once again, congratulations on your selection, and we are excited to have you as part of our team. We wish you every success in your internship at “Raksha Technologies Private Limited”.

For **Raksha Technologies Private Limited**

Anthony Raj A
Assistant Manager - HR

RAKSHA TECHNOLOGIES PVT LTD info@raksha.co.in www.raksha.co.in

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INTERNSHIP COMPLETION CERTIFICATE



ISO 9001:2015 | Cert : 727697
ISO/IEC 27001 : 2022 | Cert : 684037



Ref: RTPL/HR/2026/006

Date: 30th June 2026

TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Ms. Janavika P U** (Register No: 25CS120), a student of B. E (Computer Science) from St. Joseph's College of Engineering (Autonomous), Chennai has successfully completed her internship training program from 9th June 2026 to 30th June 2026 in the Technical Support Department of our organization.

During her tenure, we found her learnings to be sincere and satisfactory.

We wish **Ms. Janavika** all the best in her future endeavours!

For **Raksha Technologies Private Ltd.**

VARADACHARY ANAND

Digitally signed by
VARADACHARY ANAND
Date: 2026.07.04
16:19:11 +05'30'



Anand V – CEO

Authorized Signatory

RAKSHA TECHNOLOGIES PVT LTD info@raksha.co.in www.raksha.co.in

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GEO TAGGED PHOTOS

